

A computational ore microscope: per-pixel Stokes polarimetry for characterisation of UG2 ore

REEFPRIINT (developed as KHANYA) — Team Sonar

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CHALLENGE: COMPUTER VISION FOR REAL-TIME MINERALOGICAL CHARACTERISATION

The problem. PGE recovery from UG2 ore is governed by base-metal sulphide deportment: pentlandite is the principal PGE host and floats, pyrrhotite is depressed and carries little PGE, and fine chromite reporting to concentrate is a smelter penalty. Resolving that assemblage today means SEM-based automated mineralogy — QEMSCAN or MLA — accurate, but offline, slow, capital-intensive, and therefore never in the control loop. The obvious optical shortcut fails on physics rather than on engineering: hyperspectral reflectance identifies minerals by molecular absorption features, and chromite — 50–75 vol% of UG2 — is an opaque spinel with none. On chromitite, a spectrometer is a brightness meter.

Our approach. Opaque ore minerals have been identified since the 1940s by quantitative specular reflectance, bireflectance and anisotropy under crossed polars, and published automated optical mineralogy — the Castroviejo–Pirard line, CAMEVA, AMCO — classifies on multispectral reflectance alone. We add an axis reflectance does not contain: recovery of the **full linear Stokes vector at every pixel** from a rotating-analyser image series. Reflectance cannot reliably separate pentlandite (cubic, isotropic) from pyrrhotite (moderately bireflectant, anisotropic); the per-pixel Stokes signature can — and that separation governs PGE deportment and flotation response. Polarised-light *imaging* in ore microscopy is established prior art (Pirard, Lebichot & Krier, 2007); per-pixel Stokes vector recovery is not, and that narrow claim is the one we defend.

Methods and technologies. Segmentation of polished-section imagery (PyTorch, Apache-2.0 backbones), reflectance calibrated against the IMA/COM Quantitative Data File, harmonic inversion for the Stokes parameters, and conformal prediction for calibrated per-prediction uncertainty. Three decisions distinguish the build. First, **acquisition geometry is measured, not assumed**: a rotating analyser modulates at the second harmonic and yields the Stokes vector; stage rotation under crossed polars modulates at the fourth and yields only extinction depth. Fitting the Stokes model to a stage rotation returns zero anisotropy for every anisotropic grain, silently — so the inversion refuses to run until geometry is confirmed from the data. Second, **every quantity carries its provenance** — measured, cited, stipulated, assumed — through arithmetic, so an assumption cannot be laundered into a reported result. Third, **the system abstains**: at ore transitions it emits a conservative default with a stated reason rather than holding the previous setpoint. It runs fully offline on one laptop, on permissively licensed, SBOM-tracked dependencies.

Evidence to date. The Stokes inversion resolves a 40.4× separation between isotropic and anisotropic response on a synthetic phantom with analytic ground truth — a phantom result, not an ore result. Applied to public reflected-light data (LumenStone S3 v2; 29 sections, 116 000 pixels), our geometry discriminator found neither harmonic above detection threshold — so the geometry of the published “rotation sequences” is not established by this archive, and the Stokes inversion is not licensed to run on it. We report that negative finding and route to a fourth-harmonic extinction estimator rather than force the convenient inversion. We have also quantified the evaluation trap this domain invites: on synthetic data whose only signal is section identity, patch-level splitting flatters error by a factor of 126 against an honest locality-grouped split. All results run under 249 automated tests.

Expected outcomes and impact. Per-particle modal mineralogy and base-metal sulphide deportment from reflected-light optics, carrying calibrated uncertainty and an explicit refusal mechanism, feeding three operator-facing indices: fine-chromite entrainment risk, naturally-floating-gangue load for depressant dosing, and stockpile oxidation. A ~R5 000 instrument is costed as a design; no hardware is built, and no claim here depends on one.

Falsifiability. We hold a pre-registered null hypothesis: after controlling for Cr₂O₃ and pyroxene fraction, texture carries no additional predictive signal. It is tested by nested model comparison with cluster-robust inference grouped by locality and reported whichever way it resolves; if it cannot be rejected, we will say so publicly and pivot to the oxidation index.

Sources: public data only (LumenStone, IronOreRLM), IMA/COM reflectance reference data, and Craig & Vaughan (open access, MSA). No proprietary Mintek data is used. AI coding assistants supported implementation and drafting; the scientific direction and system design are the team's own, evidenced in a continuous, timestamped commit history.